

**Maximizing Aging Well Through Dance:
How Dance Supports Cognitive Functioning and Emotional Regulation in Older Adults.**

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Abstract

Older adults can benefit from activities that support both emotional well-being and cognitive health. Dance-based programs have emerged as a potential way to address both areas simultaneously. This paper examines whether dance-based interventions can support emotional regulation and cognitive functioning in aging populations by reviewing evidence from four randomized controlled trials. Across studies, dance programs consistently improved mood and enhanced multiple aspects of cognitive functioning measured through validated psychological and neuropsychological assessments. These findings suggest that dance offers multidimensional benefits, engaging emotional, cognitive, and physiological systems simultaneously. Understanding the potential of dance as an accessible, enjoyable intervention is important for promoting healthy aging and expanding the options for older adults.

Maximizing Aging Well Through Dance:

How Dance Supports Cognitive Functioning and Emotional Regulation in Older Adults.

Older adults participate in a range of activities intended to support emotional and cognitive well-being. However, age-related cognitive changes, including declines in memory, attention, and cognitive processing can present significant barriers to independence and healthy aging. Recent studies have taken this opportunity to evaluate dance as a potential tool for healthy aging.

Dance-based activity is a promising approach, which combines physical movement, executive functioning, and social engagement. Research on dance for older adults suggests specific benefits for both emotional and cognitive health. However, the strength and nature of this evidence vary depending on the population studied, the intervention format, and the outcomes measured. As a result, it remains important to clarify whether dance can meaningfully support emotional regulation and cognitive functioning, and whether findings from studies with clinical populations, such as adults with mild cognitive impairment, generalize to broader aging communities.

This paper argues that dance-based interventions help older adults maintain and improve emotional regulation and cognitive functioning, drawing on evidence from four randomized controlled trials involving dance movement therapy, dance-based aerobic programs, and ballroom dance. Firstly, we will review the research demonstrating reductions in depressive symptoms and improvements in mood, due to dance programs. Then, we will evaluate findings on improved cognitive functioning and processing through dance-based interventions. Lastly, we will synthesize the evidence which shows that dance engages multiple emotional and cognitive systems simultaneously.

Dance Increases Emotional Well-Being

Dance-based programs help older adults improve and maintain emotional regulation by reducing depressive symptoms and negative affect, supporting the broader idea that dance contributes to emotional well-being in aging.

Vankova et al. (2014) conducted a randomized controlled trial in seven Czech nursing homes to examine the impact of a 12-week dance-based program (EXDASE) on depressive symptoms in residents aged 60 and older ($N = 162$). Depressive symptoms were measured using the 15-item Geriatric Depression Scale (GDS-15), which assesses mood through yes/no responses. Results showed a significant decrease in GDS scores in the dance group ($p = .005$), whereas the control group's symptoms showed a nonsignificant trend toward worsening ($p = .081$). These findings demonstrate that dance-based activity, even at once-weekly frequency, can measurably reduce depressive symptoms in institutionalized older adults.

Ho et al. (2020) similarly reported mood improvements in their randomized controlled trial of 204 older adults with mild dementia assigned to 12 weeks of dance movement therapy (DMT). Mood outcomes were measured using the Visual Analogue Mood Scale (for positive and negative mood), a short version of the Geriatric Depression Scale, and a validated loneliness scale. The DMT group showed significant reductions in depression, loneliness, and negative mood (effect sizes $d = 0.33$ - 0.42) after the intervention. These results indicate that expressive, rhythmic, and interactive dance-based sessions likely meaningfully improved emotional states in older adults with cognitive impairment.

Both studies document mood improvements using validated geriatric depression and mood scales. Despite differences in settings (nursing homes vs. community/outpatient settings),

population (general older adults vs. older adults with dementia), and dance format, a consistent pattern emerges: participation in structured dance programs leads to measurable reductions in depressive symptoms. Both studies also used blind assessors.

Together, these findings suggest that dance reliably reduces depressive symptoms and improves mood in diverse older adult populations. The improvements measured directly through GDS and other mood scales support the argument that dance is a practical and evidence-based approach for enhancing emotional well-being.

Dance Improves Cognitive Flexibility

Empirical evidence shows that dance-based interventions enhance cognitive processing, and related cognitive skills that typically decline with age.

Sánchez-Alcalá et al. (2025) conducted a randomized clinical trial with 92 older adults with mild cognitive impairment (MCI), of whom 47 participated in a 12-week dance-based aerobic program. Cognitive outcomes were assessed across multiple domains, including executive function (TMT), global cognition (MMSE), cognitive impairment (MoCA), and verbal fluency (Isaac Semantic Fluency Test). Following the intervention, the dance group demonstrated significant improvements across all measures, including executive function ($d = 0.64$), verbal fluency ($d = 0.61$), and cognitive impairment ($d = 1.22$). These results show that dance can strengthen core executive skills such as cognitive flexibility and processing speed.

Lazarou et al. (2017) evaluated a 10-month International Ballroom Dancing program in 129 Greek older adults with amnesic MCI, comparing it to a non-dance control group. Using a comprehensive set of neuropsychological tests, including the MMSE, MoCA, Rivermead

Behavioral Memory Test, and verbal fluency tests, the authors found that the dance group showed significantly better cognitive performance after a follow-up, particularly in measures relating to memory and executive functioning.

Both studies used randomized designs and validated neuropsychological assessments. Despite differences in dance type (aerobic vs. ballroom) and program duration (12 weeks vs. 40 weeks), both trials observed improvements across various cognitive domains. This suggests that dance's cognitively demanding features, such as memorizing sequences, coordinating movements, and adjusting to rhythms, may stimulate cognitive processes.

Overall, the evidence clearly suggests that dance improves global cognition in older adults, reinforcing the hypothesis that dance helps maintain cognitive functioning with age and that it can be an effective cognitive intervention for older adults.

Dance Engages Multiple Cognitive and Emotional Systems

Dance benefits both executive function and emotional regulation because it engages cognitive, emotional, and physiological systems simultaneously, as reflected in the multi-domain improvements across studies.

Ho et al. (2020) reported that DMT improved mood at post-intervention and also enhanced daily functioning sustained at one-year follow-up. Sánchez-Alcalá et al. (2025) found improvements in executive function, verbal fluency, and general cognitive functioning following a structured dance-based intervention. Lazarou et al. (2017) observed enhancements across a broad set of neuropsychological tests, including memory, executive skills, and verbal fluency, after a long-term ballroom dance program.

Across these studies, dance produced improvements in both emotional (negative mood, loneliness, stress physiology) and cognitive domains (executive function, memory, verbal fluency). Importantly, these benefits were measured using established tools such as the GDS, mood scales, cortisol assays, and standardized neuropsychological tests. The repeated finding that dance influences multiple domains suggests that its cognitive-emotional benefits may arise from its multi-faceted nature. By requiring attention, sequencing, social engagement, and emotional expression, dance may be a unique way to promote cognitive-emotional functioning in ways that align with the observed improvements.

Conclusion

This paper examined whether dance-based interventions can support emotional regulation and cognitive functioning in older adults. Across four randomized controlled trials, the evidence demonstrated meaningful improvements in emotional well-being, including reductions in depressive symptoms and negative affect. The other findings include gains in cognitive functioning, particularly memory, verbal fluency, and processing.

More broadly, the reviewed studies suggest that the value of dance lies not in any single component but in its integration of physical movement, emotional expression, and social interaction. This multifaceted nature of dance, may help explain why improvements appear across both emotional and cognitive domains. Although much research has focused on older adults with mild cognitive impairment, the consistency of findings across different cultural settings and dance formats indicates strong potential for generalization.

Nonetheless, limitations remain. Dance programs may be less accessible to individuals with mobility restrictions or limited community resources, and selection biases may influence

who enrolls in such studies. Future research should examine diverse older adult populations, compare dance to other activities, and explore long-term engagement.

Overall, the existing evidence supports the conclusion that dance is an effective, engaging intervention that can contribute meaningfully to healthy aging.

References

- Ho, R. T. H., Fong, T. C. T., Chan, W. C., Kwan, J. S. K., Chiu, P. K. C., Yau, J. C. Y., & Lam, L. C. W. (2020). Psychophysiological Effects of Dance Movement Therapy and Physical Exercise on Older Adults With Mild Dementia: A Randomized Controlled Trial. *The Journals of Gerontology. Series B, Psychological Sciences and Social Sciences*, 75(3), 560-570. <https://doi.org/10.1093/geronb/gby145>
- Lazarou, I., Parastatidis, T., Tsolaki, A., Gkioka, M., Karakostas, A., Douka, S., & Tsolaki, M. (2017). International Ballroom Dancing Against Neurodegeneration: A Randomized Controlled Trial in Greek Community-Dwelling Elders With Mild Cognitive impairment. *American journal of Alzheimer's Disease and Other Dementias*, 32(8), 489-499. <https://doi.org/10.1177/1533317517725813>
- Sánchez-Alcalá, M., Aibar-Almazán, A., Carcelén-Fraile, M. D. C., Castellote-Caballero, Y., Cano-Sánchez, J., Achalandabaso-Ochoa, A., Muñoz-Perete, J. M., & Hita-Contreras, F. (2025). Effects of Dance-Based Aerobic Training on Frailty and Cognitive Function in Older Adults with Mild Cognitive Impairment: A Randomized Controlled Trial. *Diagnostics (Basel, Switzerland)*, 15(3), 351. <https://doi.org/10.3390/diagnostics15030351>
- Vankova, H., Holmerova, I., Machacova, K., Volicer, L., Veleta, P., & Celko, A. M. (2014). The effect of dance on depressive symptoms in nursing home residents. *Journal of the American Medical Directors Association*, 15(8), 582-587. <https://doi.org/10.1016/j.jamda.2014.04.013>